TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Continuous Unidirectional Perfusion SmartLid for Hydrostatic Microphysiological Platforms**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (X), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio-Medical		Bio-Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic devices & Diagnostic devices	Clinical chemistry & biological analysis instruments Bio-signal measurement / Diagnostic devices	Medical-Bio	Pharmaceutical bio-based technologies and systems Bio-Artificial Organs
Concept of Proposed Technology	We propose a technology enabling continuous, unidirectional perfusion of organ-onchip and microphysiological systems through an integrated, disposable SmartLid equipped with -on-lid liquid actuation (pneumatic micro-valves and micro-pumps). The lid interfaces with standard microfluidic labware for pressure-driven flow and is driven by a compact, incubator-ready- pressure/air-gap control module. The system supports multi-inlet feeding, programmed injections, recirculation, and long-term tissue culture, while eliminating bulky tubing and external pump racks. Potential uses include immune-competent assays, vascularized organoid & multi-organ models, pharmacokinetics/pharmacodynamics studies, and high-content screening.			
Rationale for Joint R&D Project	The proposed joint R&D builds on insights gained through ongoing collaborative work with Korean microphysiological system developers (on-going collaboration with Qureactor's Curiochips within the 11th Swiss-South Korean Bilateral Call for joint research projects), which has helped refine CSEM's SmartLid perfusion and recirculation concept and ensure compatibility with injection-molded organ-on-chip platforms. These exchanges have demonstrated a clear and growing market need: many Korean and global MPS providers continue to rely on hydrostatic pressure-driven chips because they offer simplicity, robustness, and low-cost operation. However, these systems are fundamentally limited by bidirectional perfusion, small on-chip volumes, and poor control over			

	dynamic flow conditions, which restricts their ability to model vascularized tissues or multi-organ interactions. Rather than replacing these widely adopted hydrostatic platforms, we propose to develop complementary tools that elevate their performance while preserving their core operating principles. By adding an integrated, compact actuation layer, the SmartLid concept transforms passive hydrostatic chips into physiologically relevant perfusion systems capable of stable unidirectional flow, recirculation, and programmable multi-inlet dosing. This approach allows end-users to maintain their existing workflows and device formats while unlocking advanced biological capabilities previously accessible only through complex external pump systems. Joint R&D is therefore essential: it combines Swiss expertise in micro-actuation, flow control, and system integration with the strengths of Korean SMEs in scalable MPS fabrication, application-driven evaluation, and market proximity. Together, these competencies support the development of a next-generation perfusion platform tailored to industry needs - enhancing the usability, physiological relevance, and commercial competitiveness of hydrostatic MPS platforms across vascularized and multi-organ applications.
Key Words	1. Continuous unidirectional perfusion 2. Microphysiological system 3. Multi-organs 4. Integrated miniaturized liquid actuation 5. Multi-inlet programmable dosing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 4	End of the Project	Level 6
R&D Objectives	**Technology state of the art from CSEM:** • SmartLid prototype with integrated membrane valves and pumps that enables stable unidirectional recirculation through the biological model (1) • POC completed through previous collaboration project (2) • On-going industrialization of the membrane integration and re-validation of the recirculation in joint project with Qureactor/Curiochips **R&D objectives for new collaborations with Korean SMEs:** • Integrate multi-inlet feeding and gradient dosing functions. • Design a compact actuation module compatible with standard incubators and minimal tubing. • Ensure compatibility with injection-molded Korean MPS devices through co-design and testing. • Validate biological relevance in vascularized and interconnected organ models. • Deliver industrialization-ready designs and manufacturable lid architectures.			
R&D Target Specifications	• Multi-inlet design and operation: at least 2 inlets, enabling continuous perfusion with/without gradient schemes • Unidirectional flow: 0.1 - 20 µL/min, stable ±10% for ≥72 h. • Shear stress: Compatible with endothelial and vascularized tissues. • Injection capability: 10 - 200 µL /gradient dosing with ≤5% CV; <10 s latency. • Pressure robustness: Leak-free operation up to ±1.0 bar. • Manufacturability: Injection-moldable lid components; multilayer sealing (e.g., laser welding).			
R&D Plan	• WP1 (M1–M6): Architecture, valve/pump design, multi-inlet manifolds, materials. • WP2 (M4–M12): Prototype SmartLid fabrication + actuation module development. • WP3 (M10–M22): Flow validation (unidirectional flow, dosing, recirculation), and biological validation with vascularized and multi-organ MPS • WP4 (M18–M24): Pilot injection-molded lids; reliability & DFM package.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Lead SmartLid actuation design, system integration, and control firmware. • Conduct flow characterization, reliability tests, and bio validation. • Provide industrialization support and manufacturability guidance. • Align developed solutions with insights from prior collaborations and Korean application needs.
Expected R&R of Korean SME Partner	• Provide requirements based on their MPS platforms and end-user workflows. • Co-design injection-moldable geometries and lid-plate interfaces. • Conduct device and biological testing using local MPS models. • Contribute to pilot production and commercialization planning.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Achievements** A fully integrated SmartLid enabling hydrostatic MPS platforms to achieve controlled unidirectional perfusion, recirculation, and programmable multi-inlet dosing. **Commercialization Potential** The solution provides a drop-in upgrade for the large installed base of hydrostatic-driven MPS devices, significantly improving their performance while maintaining user-friendly workflows. This reduces adoption barriers and expands market reach across pharma, biotech, and academic sectors. **Broader Industrial & Economic Impact** The technology strengthens the competitiveness of Korean MPS-platform/labware providers by enabling physiologically relevant culturing capabilities without redesigning their platforms. It promotes animal-use reduction, accelerates international standardization, and reinforces Swiss/Korean cooperation in advanced bioengineering.

6. References

(1) “Automated liquid exchange on passive-driven flow microfluidic devices”; EP23162537 (03/23): LIQUID HANDLING COVER FOR A MICROFLUIDIC DEVICE

(2) Smart Labware for Organoid Perfusion and 3D Cell Culture (https://www.csem.ch/en/news/smart-labware-for-continuous-organoid-on-plate/)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization